

LAB EXPERIMENT-6

Aim : To implement and verify logical shift, arithmetic shift, rotate, and rotate with carry operations (RS, LS, ARS, ALS, RR, RL, RRC, RLC) using an 8-bit shift register in Logisim.

Theory :

A shift register is a sequential digital circuit made using flip-flops that is used to store and shift binary data. In an 8-bit shift register, 8 flip-flops are connected such that each flip-flop stores one bit, and data is transferred from one stage to another on each clock pulse.

Shifting operations are essential in digital systems for data manipulation, arithmetic operations, and communication systems.

Types of Operations :

1. Logical Right Shift (RS) :

In this operation, all bits are shifted one position to the right, and the most significant bit (MSB) is filled with 0.

Ex : 10110110 → 01011011

2. Logical Left Shift (LS) :

All bits are shifted to the left, and the least significant bit (LSB) is filled with 0.

Ex : 10110110 → 01101100

3. Arithmetic Right Shift (ARS) :

Similar to logical right shift, but the MSB (sign bit) is preserved and copied during shifting. This is used for signed number representation.

Ex : 10110110 → 11011011

4. Arithmetic Left Shift (ALS) :

Same as logical left shift; bits shift left and LSB is filled with 0. It effectively multiplies the number by 2 (for unsigned numbers).

Ex : 10110110 → 01101100

5. Rotate Right (RR) :

In rotate right, bits are shifted to the right in a circular manner. The LSB is moved to the MSB position.

Ex : 10110110 → 01011011

6. Rotate Left (RL) :

Bits are shifted to the left circularly. The MSB is moved to the LSB position.

Ex : 10110110 → 01101101

7. Rotate Right with Carry (RRC) :

In this operation, the LSB is moved to the carry flip-flop, and the previous carry value is inserted into the MSB.

Ex : Assume carry=1

10110110 → 11011011

New Carry = 0

8. Rotate Left with Carry (RLC) :

The MSB is moved to the carry flip-flop, and the previous carry value is inserted into the LSB.

Ex : Assume carry=1

10110110 → 01101101

New Carry = 1

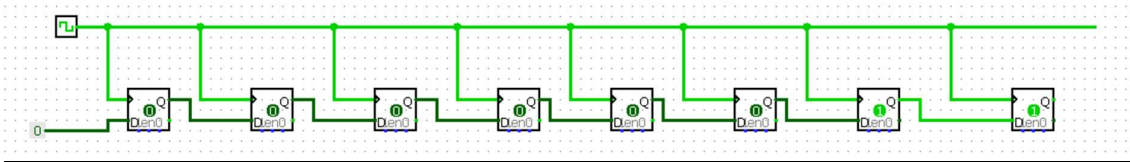
Procedure :

1. Open Logisim and create a new circuit.
2. Place 8 D Flip-Flops from the memory library and arrange them in a row to form an 8-bit shift register.
3. Label the outputs of the flip-flops as Q7, Q6, Q5, Q4, Q3, Q2, Q1, and Q0.
4. Connect a common clock signal to all flip-flops so that they operate simultaneously.
5. Connect the enable (EN) input of all flip-flops to logic HIGH (1).
6. Provide initial input by manually setting the flip-flop values (0 or 1).
7. Make connections between outputs (Q) and inputs (D) according to the required operation:
 - For Logical Right Shift (LR): shift bits to the right and insert 0 at MSB.
 - For Logical Left Shift (LS): shift bits to the left and insert 0 at LSB.
 - For Arithmetic Right Shift (ARS): shift right and copy MSB.
 - For Arithmetic Left Shift (ALS): shift left similar to logical left shift.
 - For Rotate Right (RR): connect LSB to MSB for circular shifting.
 - For Rotate Left (RL): connect MSB to LSB for circular shifting.
 - For Rotate Right with Carry (RRC): move LSB to carry and carry to MSB.
 - For Rotate Left with Carry (RLC): move MSB to carry and carry to LSB.

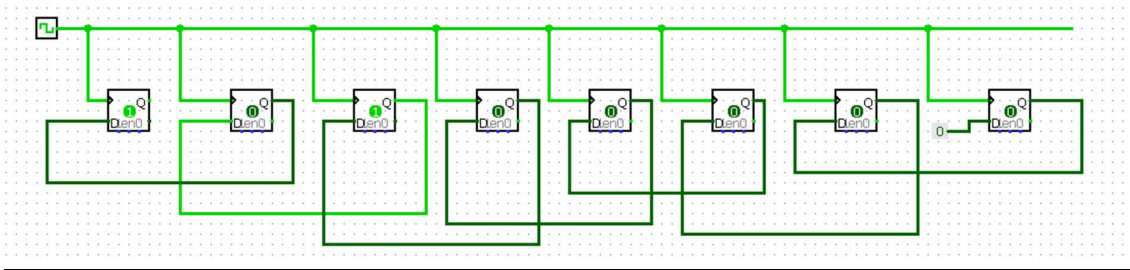
8. Apply clock pulses using the clock component.
9. Observe and record the output after each clock pulse.

Observation :

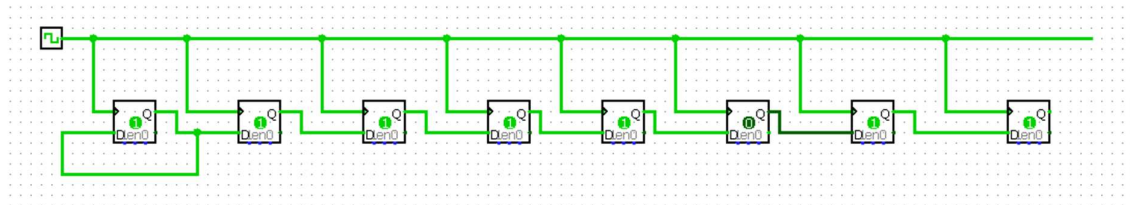
1. Logical Right Shift (RS) :



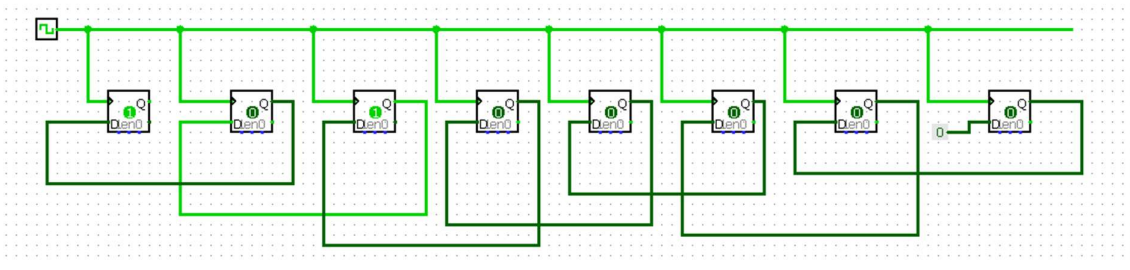
2. Logical Left Shift (LS) :



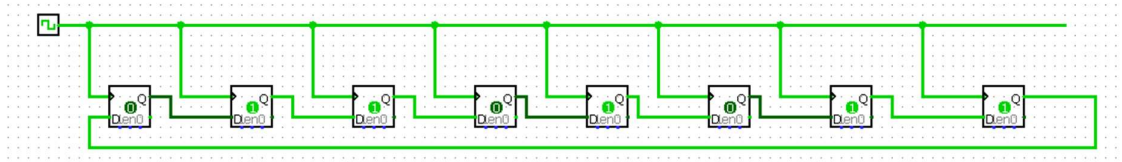
3. Arithmetic Right Shift (ARS) :



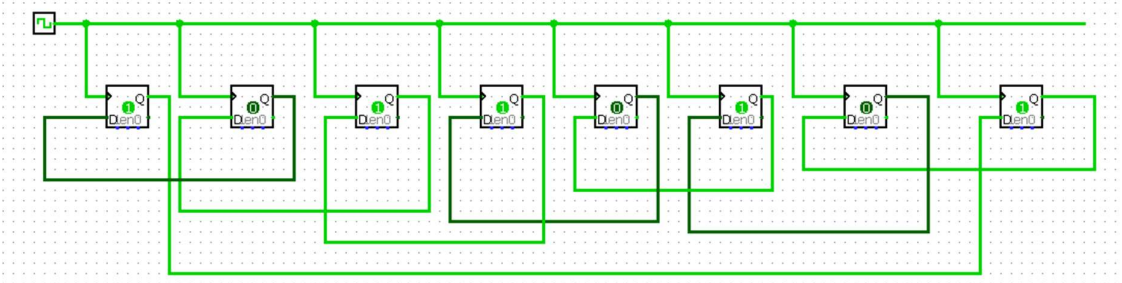
4. Arithmetic Left Shift (ALS) :



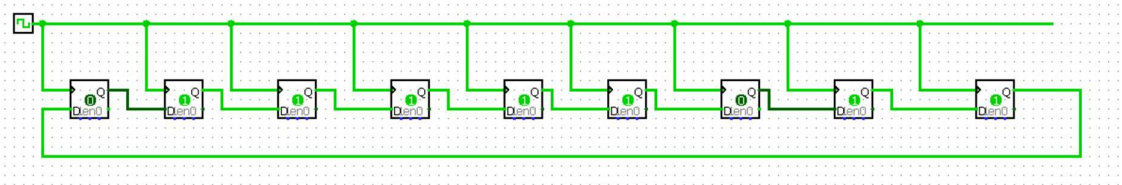
5. Rotate Right (RR) :



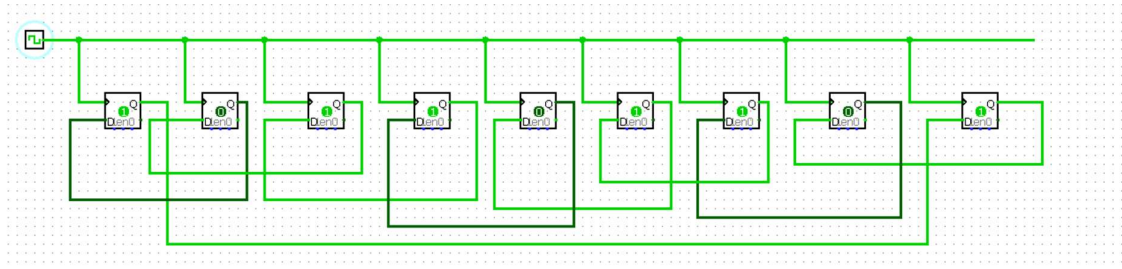
6. Rotate Left (RL) :



7. Rotate Right with Carry (RRC) :



8. Rotate Left with Carry (RLC) :



The outputs of the 8-bit shift register were observed for different operations after applying clock pulses. It was noted that bits shifted correctly in left and right directions, rotation operations produced circular movement of bits, and arithmetic right shift preserved the sign bit. The results obtained were in accordance with the expected theoretical values.

Conclusion :

The experiment was successfully carried out using an 8-bit shift register in Logisim. Various operations such as logical right and left shift, arithmetic shift, rotate left and right, and rotate with carry were implemented and tested. The results obtained after applying clock pulses matched the expected theoretical values. This experiment helped in understanding the working of shift registers, data movement, and the role of clock and control logic in sequential circuits.